Widely Used Machine Learning Models in Industry: Trends and Outlook

Modern industrial and organizational applications of machine learning are dominated by a few well-established model families. **Surveys and industry reports consistently show that simpler, "classical" algorithms remain the most widely used** in practice, even amidst rapid progress in deep learning. This report reviews which models are most prevalent globally today – drawing on data from sources like Kaggle's annual surveys and industry studies – and then discusses how this distribution might shift in the next 5–10 years. Key trends are summarized in the table below for reference.

Model Family	Prevalence among Practitioners	Typical Use Cases / Domains
Linear/Logistic Regression	~80% use (most common)[1]	General-purpose, tabular data (classification, regression)
Decision Trees & Random Forests	~70–75% use (close 2nd)[1]	Tabular data, interpretable models (e.g. risk modeling)
Gradient Boosting Machines (e.g. XGBoost)	~60–70% use (top 3)[1]	Tabular data, high-accuracy ensemble methods (often Kaggle/competition favorites)
Convolutional Neural Networks (CNNs)	~43% use[2]	Computer vision, image recognition, some text applications
Recurrent Neural Networks (RNNs)	~30% use[2]	Sequence data (time-series forecasting, legacy NLP tasks)
Transformer-Based Models (LLMs)	~15% use in 2020[3] (rapidly growing)	Natural language processing (text, chatbots); generative AI (emerging mainstream)

Note: Percentages above are approximate and based on multi-select survey data (Kaggle, 2020) indicating the proportion of data scientists who reported using each model type in their work.

Current Industry Adoption of ML Model Families

Dominance of Classical Models. In real-world industrial settings, **"classical" machine learning algorithms (linear models and decision-tree-based models) are by far the most widely employed**. According to Kaggle's 2020 *State of Data Science & ML* survey, the single most popular algorithm was linear regression, which **over 80% of data scientists reported using**[1]. This includes linear and logistic regression models, often used as interpretable and reliable tools for tasks like risk scoring, customer analytics, and forecasting. Close on its heels, **decision tree methods – including simple decision trees**

and ensemble variants like Random Forests – were the second most-used algorithms, with usage rates not far behind the linear models[1]. In the same survey, **gradient boosting machine (GBM) algorithms** (e.g. XGBoost, LightGBM) came in as the third most popular category[1]. These GBM models have become go-to solutions in many structured data competitions and business applications due to their high predictive performance. In Kaggle's 2021 report, this pattern remained clear: *"the most commonly used algorithms were linear and logistic regression, followed closely by decision trees and random forests,"* with gradient boosting machines being the next most popular advanced method[4]. Practicing data scientists often confirm this trend anecdotally – for many problems, a well-tuned **XGBoost or a simple logistic regression tends to perform remarkably well**, leading one practitioner to quip that *"XGBoost and Linear/Logistic regression are all you need… in practice"* for most workaday modeling tasks[5]. In summary, **regression and tree-based models (including ensembles) form the backbone of most industrial AI systems today**, due to their robustness, speed, and ease of use/interpretation in business contexts[1].

Role of Deep Learning in Practice. While deep learning grabs headlines, its **actual usage in industry, outside of specific domains, is more selective**. Kaggle's surveys show that a significant minority of practitioners do use neural network models, but not to the same ubiquitous extent as the simpler models. For example, in 2020 about **43% of data scientists had used convolutional neural networks (CNNs)** – making CNNs the most widely used deep learning approach, primarily for computer vision and image-related applications[2]. Other neural architectures saw lower but notable use: **RNNs (recurrent neural networks)** were used by ~30% (for sequence data, time series, and some older NLP tasks), and **Transformer-based models** (the class underpinning modern large language models) were used by roughly 15% of practitioners in 2020[2]. This indicates that by 2020, only a small portion of the community was working directly with Transformers or large language models (LLMs) in their workflows. However, **this portion has grown rapidly in recent years**. By 2022, usage of Transformers had increased substantially – one analysis found that among machine learning engineers (a subset of survey respondents), **38% were using transformer models by 2022**, reflecting the surge of interest in NLP and LLMs[6]. Deep learning is thus increasingly common, but it tends to be **concentrated in specific domains**: e.g. CNNs are standard in image recognition, object detection, and other vision tasks; recurrent or transformer models are employed for natural language processing, speech recognition, and recommendation systems at tech companies. Meanwhile, for many classic business analytics problems involving **tabular data (spreadsheets, databases)**, companies often still favor the simplicity of linear models or decision-tree ensembles. Indeed, Kaggle's 2022 survey data confirms that **data scientists most regularly rely on linear/logistic regression, tree ensembles, and gradient boosting**, whereas deep neural networks are leveraged primarily when the problem domain (like vision or text) warrants it[7]. This distribution suggests that despite the rapid advancements in AI, **the industry at large has been relatively conservative in adopting new model classes** – embracing deep learning where it clearly outperforms, but continuing to utilize tried-and-true algorithms for the many problems where they are effective.

Outlook for the Next 5–10 Years

Looking ahead, we can expect **shifts in the balance of model usage** as technology and industry practices evolve, though classical methods will likely *not* disappear. In the next 5–10 years, **classical models such as linear/logistic regression and decision trees will probably remain a staple** of industrial ML. There are strong reasons for this: these models are well-understood, computationally efficient, and often easier to interpret and deploy at scale. Many organizations (especially in regulated sectors like finance or healthcare) value the transparency of simpler models. Thus, for structured data tasks – credit scoring, churn prediction, supply chain forecasting, etc. – it's reasonable to predict that a large share of solutions will still involve linear models or tree ensembles in five years. **However, their relative dominance may diminish** as more powerful techniques become accessible. The trend so far has been that as soon as advanced models (like GBMs or neural nets) prove their worth and become easier to use, they get integrated into the mainstream. We have already seen gradient boosting overtake single decision trees in popularity, and we may see similar inroads from deep learning. In particular, **autoML tools and cloud platforms are increasingly abstracting the modeling process**, potentially introducing more complex models under the hood. This means even some users of classical models might unknowingly start benefiting from neural networks or hybrid models if those yield better accuracy for a given task.

Rising Influence of Deep Learning and LLMs. It is in the realm of unstructured data and content generation that the **biggest shifts are likely to occur.** Deep learning approaches – especially **large pre-trained models** – are poised to gain a much larger share of industrial use. The clearest evidence is the explosive adoption of **generative AI and large language models in the past couple of years**. By late 2023, an industry survey by O'Reilly found that *two-thirds of organizations were already using generative AI* in some form[8] – a stunning adoption rate for a technology barely one year into mainstream awareness. This unprecedented pace suggests that **LLM-based applications (such as GPT-powered chatbots, writing assistants, and code generators) will become commonplace in many companies**. Over the next 5–10 years, we can expect NLP-heavy tasks (customer service bots, document analysis, marketing content creation, etc.) to shift strongly toward transformer-based models and other deep learning techniques, overtaking earlier approaches like basic NLP pipelines or manual rules. **Domains that involve image, audio, and video data have already largely embraced deep learning** (for example, virtually all modern image recognition systems rely on CNN or transformer architectures). These domains will continue to evolve with new model architectures (e.g. vision transformers, multimodal models) and will likely see even deeper penetration of AI into operations (e.g. automated quality inspection via vision, voice-activated enterprise assistants, etc.). Another domain expecting a strong shift is **business intelligence and knowledge management** – here large language models can be fine-tuned to answer company-specific questions or generate insights from text data, something classical models could not do. In summary, **the share of deep learning models in production is set to increase significantly**, driven by the clear performance advantages on unstructured data and the growing ease of deploying these models via cloud services and pre-trained APIs.

That said, **classical methods will remain relevant and even dominant in certain areas** for the foreseeable future. It's likely that in 5–10 years, the *overall* landscape will be more mixed: a typical large organization might use simple regression or tree models for many of its smaller-scale predictive tasks, while also using deep learning for specific high-value applications (like an image analysis pipeline or an NLP-powered analytics tool). The **strongest shifts will occur in fields that can readily exploit the abundance of data with deep learning** – for example, **customer engagement and CRM** can leverage LLMs for personalized communication; **healthcare diagnostics** will further adopt deep neural networks for medical imaging and possibly patient record analysis; **manufacturing** may increasingly use computer vision (CNN/transformer models) for quality control and robotics. Conversely, in domains where data are tabular or limited in size – e.g. small businesses with modest databases, or problems requiring strict interpretability – we may find that **linear models and decision trees continue to be the workhorses** due to their proven reliability and ease of explanation. Importantly, we might also see **hybrid approaches** become common: for instance, using large neural networks to generate features or embeddings from unstructured data, which are then fed into simpler models for decision-making. This could blur the line between "classical" and "advanced" techniques and gradually introduce more deep learning components into traditionally linear-model domains.

In conclusion, **classical ML models currently lead in real-world usage, but the gap is closing as deep learning and especially LLM-based methods rapidly gain ground.** Today, the global industry leans on linear/logistic regression, decision trees, random forests, and gradient boosting as the familiar, widely deployed solutions[4][1]. Deep learning models are used strategically – dominantly in vision and language tasks – and their adoption is accelerating alongside breakthroughs in AI. Over the next 5–10 years, we can reasonably predict a more balanced paradigm: **classical models will remain integral (due to their simplicity and certain advantages), yet deep learning techniques will claim a larger share of new and evolving applications**. The strongest shifts will align with data type and domain – with **NLP and multimodal applications driving the surge of LLMs and advanced neural networks**, while many data-centric business problems continue to rely on the *"better understood"* algorithms. Organizations that stay abreast of these trends will likely maintain a toolkit that spans both ends of the spectrum, applying the right model family to each problem: from a straightforward regression for an interpretable financial risk model, to a giant pre-trained transformer for an automated customer support agent. The net outcome is a future where **machine learning in industry becomes even more pervasive and powerful**, leveraging classical and deep learning models each where they add the most value.